panMP and mpxForPan on the 193500 recording

MATLAB R2026a and Python were given the same samples and the same arguments.

Input

File: SOCAL_H_76_230426_193500_df100.x.x.wav. Sample rate 2000 Hz. The file contains 72,000,000 samples (10.0 hours), mono int16.

Both programs received the first 16,000 samples, converted to float64, which is the first 8.000 seconds. A full pairwise profile of all 72 million samples is not practical: each motif length compares every window with every later window.

Arguments

panMP(x). MATLAB hardcodes motif lengths 32, 96, 160, 224, 288, 352, 416, and 480 (step 64, through 512) and calls mpxForPan(x, i-2, i). Python panMP(x) uses those defaults. The figure was not shown.

mpxForPan(x, minlag, window) was also called directly with (30, 32), (222, 224), and (478, 480).

This MATLAB install has no Statistics Toolbox. zscore was (x - mean(x)) / std(x) with the sample standard deviation, the same definition used in panMP.py.

mpxForPan

minlag	window	profile length	max |distance difference|	index
30	32	15969	5.329e-15	match after +1
222	224	15777	5.684e-14	match after +1
478	480	15521	6.395e-14	match after +1

Python indices are 0-based. Adding 1 makes them equal to the MATLAB indices at every position.

panMP

Check	Result
Surface shape	15456 by 543, both
NaN columns	same positions
max |finite difference|	2.331e-15

Result

On this recording, with these arguments, the Python functions and the MATLAB functions return the same profiles and the same pan surface.